Anomalous Criticality in the Electrical Resistivity of $La_{2-x}Sr_xCuO_4$

R. A. Cooper,[1] Y. Wang,[1] B. Vignolle,[2] O. J. Lipscombe,[1] S. M. Hayden,[1] Y. Tanabe,[3] T. Adachi,[3] Y. Koike,[3] M. Nohara,[4]* H. Takagi,[4] Cyril Proust,[2] N. E. Hussey[1]†

The presence or absence of a quantum critical point and its location in the phase diagram of high-temperature superconductors have been subjects of intense scrutiny. Clear evidence for quantum criticality, particularly in the transport properties, has proved elusive because the important low-temperature region is masked by the onset of superconductivity. We present measurements of the low-temperature in-plane resistivity of several highly doped $La_{2-x}Sr_xCuO_4$ single crystals in which the superconductivity had been stripped away by using high magnetic fields. In contrast to other quantum critical systems, the resistivity varies linearly with temperature over a wide doping range with a gradient that scales monotonically with the superconducting transition temperature. It is maximal at a critical doping level (p_c) ~ 0.19 at which superconductivity is most robust. Moreover, its value at p_c corresponds to the onset of quasi-particle incoherence along specific momentum directions, implying that the interaction that first promotes high-temperature superconductivity may ultimately destroy the very quasi-particle states involved in the superconducting pairing.

An important theme in strongly correlated electron systems is quantum criticality and the associated quantum phase transitions that occur at zero temperature upon tuning a nonthermal control parameter, g (e.g., pressure, magnetic field H or composition), through a critical value, g_c. One feature of such a system is the influence that critical fluctuations have on the physical properties over a wide region in the (T, g) phase diagram above the quantum critical point (QCP), inside which the system shows marked deviations from conventional Landau Fermi-liquid behavior. A number of candidate non–Fermi-liquid systems have emerged, particularly in the heavy fermion family (*1*), although there are others, for example, certain transition metal oxides (*2*), that display similar characteristics.

The physics of copper-oxide high-temperature superconductors may also be governed by proximity to a QCP. The generic temperature-doping (T, p) phase diagram resembles that seen in the heavy fermions, with an apparent funnel-shaped region that either pierces or skirts the superconducting dome (*3*). Above this region, cuprates display an in-plane resistivity, ρ_{ab}, that varies linearly with temperature over a wide temperature (*4*) yet narrow doping (*5*) range. This T-linear resistivity has been widely interpreted, in tandem with other anomalous transport properties (*6*), as a manifestation of scale-invariant physics borne out of proximity to the QCP. This viewpoint has remained untested, largely because of the high upper critical field H_{c2} values in high-T_c cuprates that restrict access to the important limiting low-temperature region below $T_c(p)$. We used a combination of persistent and pulsed high magnetic fields to expose the normal state of $La_{2-x}Sr_xCuO_4$ (LSCO) over a wide doping and temperature range and studied the evolution of $\rho_{ab}(T)$ with carrier density, from the slightly underdoped (p = 0.15) to the heavily overdoped (p = 0.33) region of the phase diagram. Our analysis reveals the presence of a singular doping concentration in LSCO at which the electronic response changes, although in a manner distinct from that observed in other candidate quantum critical systems.

In-plane resistivity of $La_{2-x}Sr_xCuO_4$. A series of high-field $\rho_{ab}(T, H)$ measurements were carried out on overdoped LSCO single crystals with doping levels of p = 0.18, 0.21, and 0.23 (labeled hereafter LSCO18, LSCO21, and LSCO23, respectively) with the field aligned perpendicular to the CuO_2 planes in order to suppress the superconductivity. Figure 1A shows the $\rho_{ab}(T, H)$ data obtained on LSCO23. In order to track the temperature dependence of the zero-field resistivity $\rho(T, 0)$ below T_c, we used a simple, transparent technique to extrapolate the high-field $\rho_{ab}(T, H)$ data to the zero-field axis (Fig. 1B). The resultant $\rho(T, 0)$ values, plotted in Fig. 1C together with the zero-field $\rho_{ab}(T)$ curve below 70 K, are found to exhibit a T-linear dependence down to 1.5 K. For comparison, we also plotted the absolute values of $\rho(T, 48)$ at a fixed high field of 48 T obtained directly from the vertical dashed line in Fig. 1A. The temperature dependence of the latter (analysis-free) values is identical to that of $\rho(T, 0)$ and is consistent with earlier 60-T data taken on LSCO22 (*7*), showing that the analysis itself has not introduced any additional, artificial temperature dependence in $\rho(T, 0)$. Similar pulsed-field measurements and analysis were carried out for the two other doping levels as summarized in fig. S1.

Figure 2 shows the resultant $\rho(T, 0)$ values plus zero-field $\rho_{ab}(T)$ data for seven different concentrations ranging from optimal doping (p = 0.17) to the heavily overdoped, nonsuperconducting region (p = 0.33). The gradual crossover in the temperature dependence of $\rho_{ab}(T)$, from quasi-linear for LSCO17 to approximately quadratic for LSCO33, is evident in the raw data and is consistent with previous studies carried out above T_c (*5*, *8*, *9*). At low temperatures, however, $\rho_{ab}(T)$ develops predominantly T-linear behavior for the entire doping range $0.18 \leq p \leq 0.29$ [for p = 0.17, data exists only above $T_c(H = 0)$]. Although evidence for a low-T T-linear resistivity has emerged for single doping concentrations in both electron- (*10*) and hole-doped (*11*, *12*) cuprates, our measurements show that the low-T linearity in fact persists over a broad range of doping.

Single-component analysis. In heavy fermion systems, $\Delta\rho(T)$, the T-dependent part of $\rho(T)$, is often described by a single term $\alpha_n T^n$ whose exponent $n(T, H)$ evolves from the Fermi-liquid value n = 2 to some anomalous value less than 2 over a narrow temperature and magnetic field window (*13*–*15*). The anomalous exponent in $\Delta\rho(T)$ persists to low temperatures only at the critical field, H_c. In Fig. 3, we plotted a comparative $n(T, p) = d(\ln\Delta\rho)/d(\ln T)$ for LSCO by using the resistivity curves shown in Fig. 2.

For T > 50 K, the resultant phase diagram resembles that seen in prototypical quantum critical systems, with a narrow region in which $\rho_{ab}(T)$ is approximately (although not strictly) T-linear separated from a region where $\rho_{ab}(T)$ varies approximately as T^2. As the temperature is lowered, however, the situation becomes markedly different. Rather than collapsing to a single (critical) point, the T-linear region in LSCO fans out and dominates the low-T response. Intriguingly, this T-linear regime (or more precisely, the region where n < 1.1) is coincident with both the T_c parabola (long-dashed white line) and the superconducting fluctuation regime (short-dashed white line) and has thus been obscured until now by the veil of superconductivity.

Dual-component analysis. Previously, $\Delta\rho_{ab}(T)$ in overdoped, hole-doped cuprates has been expressed either as above, that is, as $\alpha_n T^n$ ($1 \leq n \leq 2$) (*16*), or as the sum of two components, $\alpha_1 T + \alpha_2 T^2$ (*11*, *17*, *18*). In fig. S2, we describe in detail why the latter is in fact the more appropriate expression for LSCO. In Fig. 4, A and B, we show the doping dependences of α_1 and α_2, respectively, for two different fitting protocols. The solid squares are coefficients obtained

[1]H. H. Wills Physics Laboratory, University of Bristol, Tyndall Avenue, Bristol, BS8 1TL, UK. [2]Laboratoire National des Champs Magnétiques Pulsés (LNCMP), UMR CNRS-UPS-INSA 5147, Toulouse 31400, France. [3]Department of Applied Physics, Graduate School of Engineering, Tohoku University, 6-6-05 Aoba, Aramaki, Aoba-ku, Sendai 980-8579, Japan. [4]Department of Advanced Materials Science, Graduate School of Frontier Science, University of Tokyo, Kashiwa-no-ha 5-1-5, Kashiwa-shi, Chiba 277-8651, Japan.

*Present address: Department of Physics, Okayama University, Tsushima-naka, Okayama 700-8530, Japan.

†To whom correspondence should be addressed. E-mail: n.e.hussey@bristol.ac.uk

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from least-square fits of the $\rho_{ab}(T)$ curves for $T \leq 200$ K to the expression $\rho_{ab}(T) = \alpha_0 + \alpha_1 T + \alpha_2 T^2$, whereas the solid circles are obtained from fits over the same temperature range to a parallel-resistor formalism $1/\rho_{ab}(T) = 1/(\alpha_0 + \alpha_1 T + \alpha_2 T^2) + 1/\rho_{max}$, where ρ_{max} (=900 ± 100 μohm•cm) represents a maximum (saturation) resistivity value (*19*, *20*). The inclusion of ρ_{max} helps to account smoothly for the escalation of $\rho_{ab}(T)$ to higher temperatures and to make the values of α_1 and α_2 insensitive to the temperature range of fitting. [For a discussion of the physical meaning of ρ_{max} and its relevance to the data, please refer to the supporting online material (SOM).] For all samples, the parallel-resistor fits (blue dashed curves in Fig. 2) are nearly indistinguishable from the $\rho_{ab}(T)$ curves over a fitting range that extends (with the exception of LSCO17) over 2 decades in temperature and a factor of 2 in doping. The open symbols in Fig. 4 relate to coefficients obtained from corresponding fits to earlier zero-field $\rho_{ab}(T)$ data (*5*) for comparison.

For both sets of data and analysis, α_1 is seen to grow rapidly with decreasing p, attaining a maximum value of around 1 μohm cm/K at p_c = 0.185 ± 0.005. In the parallel-resistor fit, α_2 remains finite at all doping concentrations and essentially p-independent down to the same doping level, at which point it rises sharply. Without ρ_{max} in the fitting procedure, α_2 becomes negligible around p = 0.20. It is important to note that none of the main findings of this study—the ubiquity of the T-linear term, the nondivergence of α_2, the

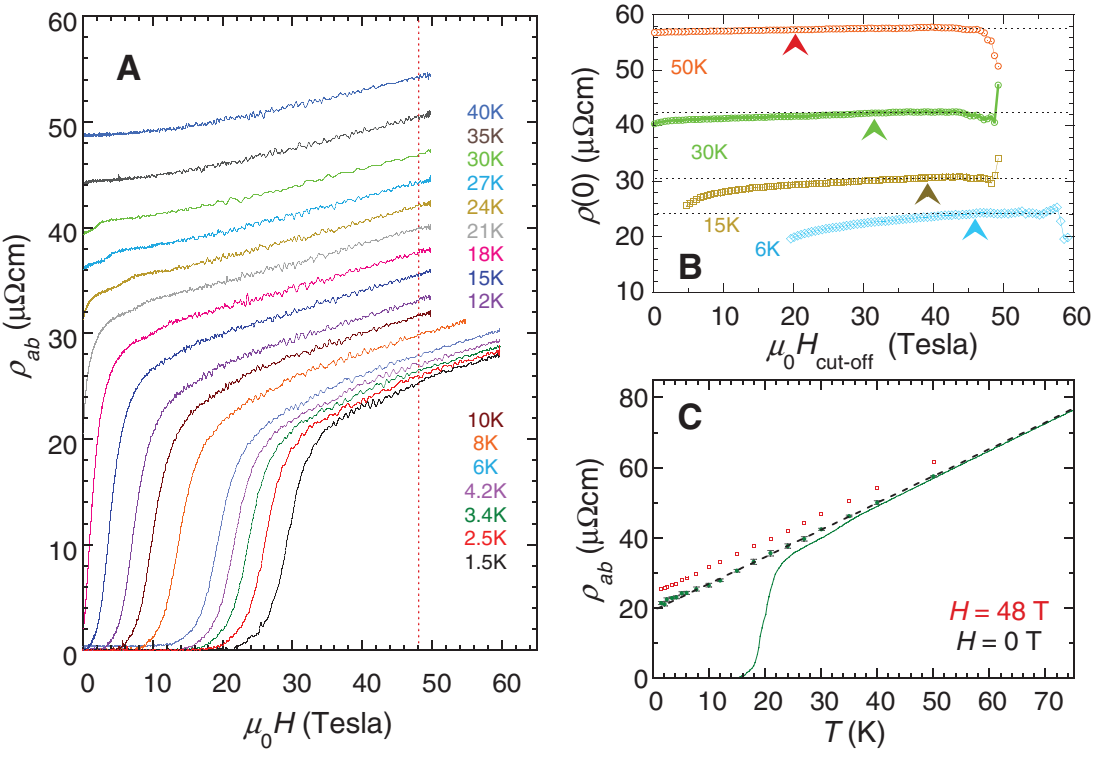


Fig. 1. Electrical resistivity of overdoped $La_{2-x}Sr_xCuO_4$ at low temperatures. (**A**) In-plane resistivity ρ_{ab} of LSCO23 as a function of temperature and magnetic field. The vertical dashed line indicates the temperature variation of ρ_{ab} at 48 T. (**B**) Mode of extraction of the zero-field resistivity $\rho(H = 0)$ for LSCO23 at each temperature. The primary data were fitted to the form $\rho(H) = \rho(0) + AH^2$ [as argued in the supporting online material (SOM) text] between some lower bound $H_{cut\text{-}off}$ and the maximum field strength used, and the deduced values of $\rho(0)$ are shown plotted as a function of these $H_{cut\text{-}off}$ values, the sample held at the various temperatures indicated. For $H_{cut\text{-}off}$ less than some indicated critical field, defined as H_{c2}, the extracted $\rho(0)$ values rise monotonically with increasing cut-off field. When $H_{cut\text{-}off}$ becomes of order H_{c2}, $\rho(0)$ reaches a plateau value. Lastly, as the field range for fitting becomes too narrow, the extrapolated value of $\rho(0)$ begins to oscillate wildly. For each temperature, $\rho(0)$ is identified by its value in the plateau region. (**C**) Zero-field $\rho_{ab}(T)$ of LSCO23 (solid green line) plotted with the extrapolated values of $\rho(0)$ (green solid squares) and of $\rho(\mu_0 H = 48\ \text{T})$ (red open squares) for the various temperatures indicated. The dashed line is a guide to the eye.

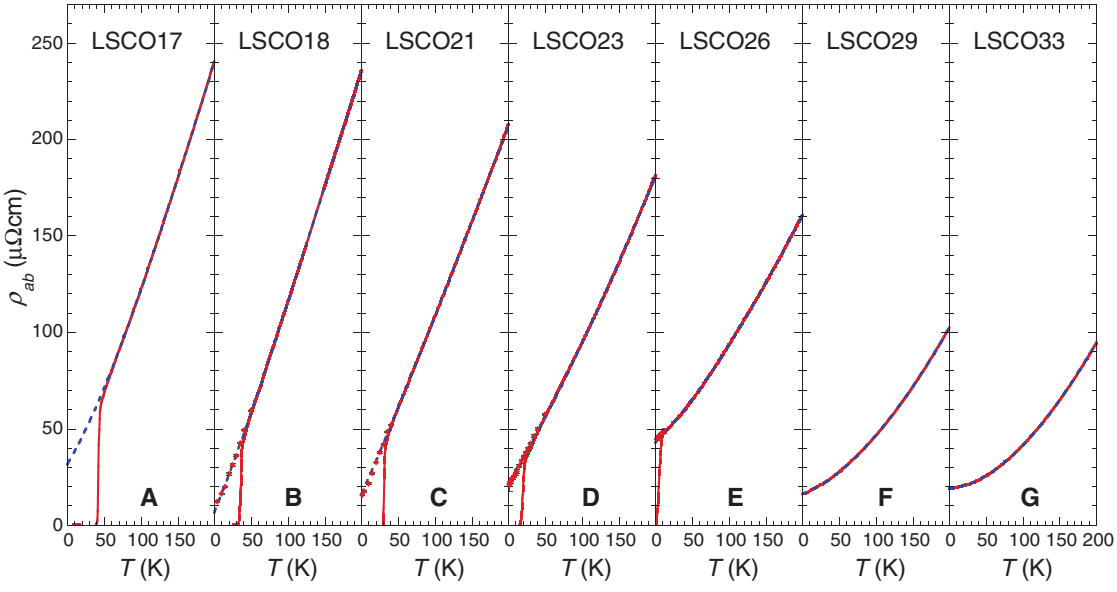


Fig. 2. Fitting of the in-plane resistivity of overdoped LSCO. Solid red lines are measured zero-field $\rho_{ab}(T)$ for (**A**) LSCO17, (**B**) LSCO18, (**C**) LSCO21, (**D**) LSCO23, (**E**) LSCO26, (**F**) LSCO29, and (**G**) LSCO33; whereas the red diamonds are corresponding extrapolated $\rho(H = 0)$ values. The blue dashed lines represent fits to the data below 200 K using the expression $1/ = 1/(\alpha_0 + \alpha_1 T + \alpha_2 T^2) + 1/\rho_{max}$ where ρ_{max} = 900 μohm•cm.

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position of the kink, nor the value of α_1 at $p = p_c$—are dependent on the type of dual-component analysis used.

The coexistence of two distinct components to $\rho_{ab}(T)$ is suggestive of some form of two-fluid model, although the fact that the two components add in the resistivity (i.e., in series) rather than in the conductivity (in parallel) implies that the two subsystems would have to coexist on a microscopic scale. Indeed, it is well documented that cuprates possess substantial microscopic inhomogeneity (*21*), and overdoped LSCO in particular is prone to microscopic phase separation into superconducting and non-superconducting regions with a physical extent on the order of the superconducting coherence length (*22*, *23*). One might then be tempted to attribute the *T*-linear component to the superconducting region (because its value scales roughly with T_c) and the quadratic term to that portion of the sample with an effective carrier density $p > 0.27$.

An alternative approach, however, is to assign the two additive coefficients to two distinct, independent quasi-particle scattering processes that coexist on the cuprate Fermi surface. Such a picture was recently proposed on the basis of angle-dependent magnetoresistance (ADMR) measurements on overdoped $Tl_2Ba_2CuO_{6+\delta}$ (Tl2201) that revealed that the scattering rate was composed of two distinct terms, a T^2 scattering term that is isotropic within the basal plane and a *T*-linear scattering rate that is strongly anisotropic, exhibiting a maximum near the Brillouin zone boundary, where the pseudogap is maximal and vanishing along the zone diagonal (*18*). In a subsequent doping-dependent study, it was intimated that this anisotropic *T*-linear term correlates with T_c, whereas the quadratic term remains constant as a function of p (*24*). Both of these trends in Tl2201 now appear to be confirmed in LSCO but over a much wider range of doping.

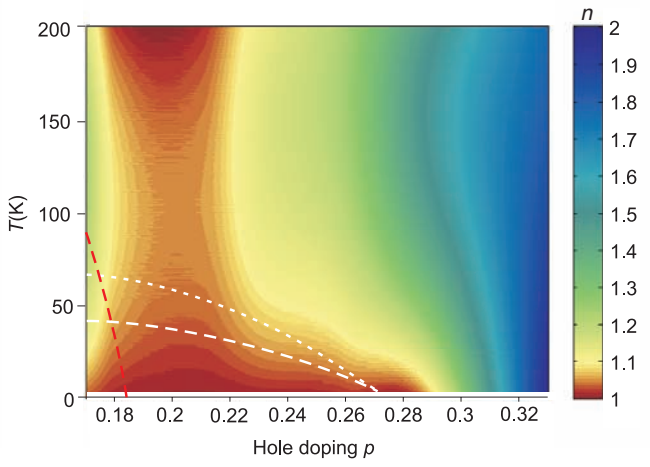


Fig. 3. Schematic of the evolution of the exponent *n* with temperature and doping in LSCO. The phase diagram is obtained directly by interpolating plots of $d[(\ln\rho_{ab} - \alpha_0)]/d(\ln T)$ versus *T* for all the resistivity curves shown in Fig. 2. In marked contrast to other quantum critical systems, the *T*-linear regime is found to grow wider with decreasing temperature. This unusual expansion of the *T*-linear region at low temperatures coincides with the superconducting dome (long dashed white line) and the region where superconducting fluctuations become significant (short dashed white line). The red dashed line represents $T^* = E_g/2$ (*36*). As stated in the text and argued in more detail in the SOM, this form of $\Delta\rho(T)$ is shown here only for comparison with the heavy-fermion compounds, but it is not the most appropriate means of describing $\Delta\rho(T)$ in LSCO or indeed other cuprates (*18*).

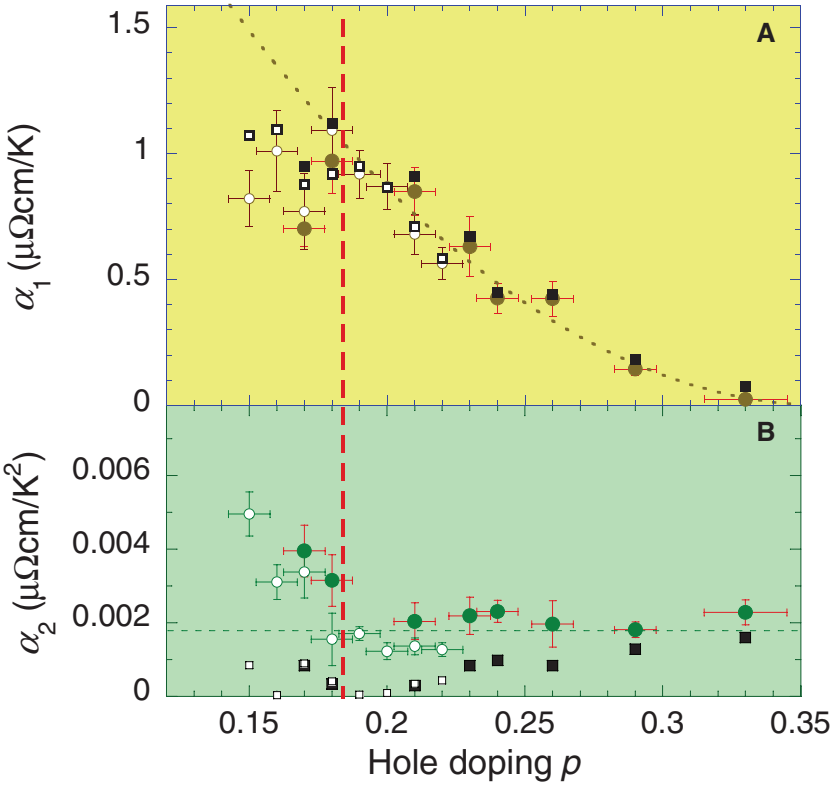


Fig. 4. Doping evolution of the temperature-dependent coefficients of $\rho_{ab}(T)$. (**A**) Doping dependence of α_1, the coefficient of the *T*-linear resistivity component. (**B**) Doping dependence of α_2, the coefficient of the T^2 resistivity component. In both panels, solid squares are coefficients obtained from least-square fits of the $\rho_{ab}(T)$ curves for $T \leq 200$ K to the expression $\rho_{ab}(T) = \alpha_0 + \alpha_1 T + \alpha_2 T^2$, whereas the solid circles are obtained from fits over the same temperature range to a parallel-resistor formalism $1/\rho_{ab}(T) = 1/(\alpha_0 + \alpha_1 T + \alpha_2 T^2) + 1/\rho_{max}$ with $\rho_{max} = 900 \pm 100$ μohm cm. The open symbols are obtained from corresponding fits made to the $\rho_{ab}(T)$ data of Ando *et al.* (*5*) between 70 K and 200 K. The dashed lines are guides to the eye. The error bars are a convolution of standard deviations in the values of α_1 and α_2 (1σ) for different temperature ranges of fitting plus systematic uncertainty in the absolute magnitude of ρ_{max}.

Anomalous criticality in LSCO. Critical fluctuations are a definitive signature of a QCP. In nearly ferromagnetic (FM) or antiferromagnetic (AFM) metals, for example, critical spin fluctuations give rise to physical properties with anomalous temperature dependences whose critical exponents depend on the nature of the spin fluctuations (FM or AFM) and on the dimensionality. For two-dimensional (2D) AFM spin fluctuations, a *T*-linear resistivity is expected that extends to very low temperatures at or near the QCP (*25*). Away from the critical point, the resistivity shows a crossover to a Fermi-liquid–like T^2 resistivity as *T* decreases. As the critical point is approached, the crossover temperature T_F decreases, and the coefficient of the T^2 resistivity diverges. Such behavior is typified by the heavy-fermion compound $YbRh_2Si_2$, where $\Delta\rho$ is found to be strictly *T*-linear over 2 decades in temperature at a critical magnetic field, H_c (*13*). Either side of H_c, $\Delta\rho$ is proportional to T^2 with a coefficient α_2 (T_F) that diverges (falls linearly) as *H* approaches H_c. Similar behavior is exhibited in the quasi-2D heavy-fermion compound $CeCoIn_5$, where the QCP coincides with the upper critical field H_{c2} (*14*), and in the bilayer ruthenate $Sr_3Ru_2O_7$ (*2*), where the QCP coincides with a metamagnetic transition.

There are three aspects of the resistivity behavior in LSCO that conflict with this conventional quantum critical picture: (i) the two

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T-dependent coefficients appear to coexist for all *T* and *p*, rather than simply merge into one another; (ii) the coefficient of the T^2 term does not diverge—it either remains constant or diminishes as one approaches p_c from the overdoped side; and (iii) the *T*-linear scattering term in LSCO persists at low *T* over an extended doping range. This is evident not only in the evolution of α_1 (Fig. 4A) but also in the single-exponent plot (Fig. 3). This extended regime of *T*-linear resistivity is a possible manifestation of a second tuning parameter other than doping, that is, a third axis in the phase diagram (e.g., magnetic field or disorder) along which the QCP may be located. In $YbRh_2Si_2$, for example, alloying with Ge broadens the field range over which the *T*-linear resistivity extends to low *T* (*13*). Alternatively, the breadth of the critical region in LSCO suggests that the strange metal physics in cuprates is associated not with a QCP but with a novel, extended quantum phase, as seen, for example, in the itinerant ferromagnet MnSi beyond a critical pressure (*26*).

Discussion. The marked kinks in α_1 and α_2 at $p_c = 0.185 \pm 0.005$ are the most revealing findings of this study. The value of p_c coincides with that inferred from a wealth of other experimental probes (*3*) and is associated with the opening of the pseudogap, an anisotropic (nodal) gap in the normal state excitation spectrum. Below p_c, a new energy (E^*) or temperature (T^*) scale emerges, which grows sharply with decreasing doping and appears to extrapolate to a value about equal to the AFM exchange energy, *J*. There are two competing scenarios describing the effect of the pseudogap on the Fermi surface in cuprates: one involving Fermi surface reconstruction into hole and electron pockets (*27*) and the other involving Fermi surface degradation into a series of disconnected Fermi patches or arcs, centered about the zone diagonals, with an arc length that is progressively reduced upon underdoping (*28*). Although the actual doping dependence of the pseudogap has been hotly debated (*3*), the consistent and systematic analysis reported here appears to confirm that there is indeed a fundamental change in the transport properties of cuprates at this specific, well-defined doping level sandwiched between optimal doping and the upper edge of the superconducting dome (*16*, *29*).

A particular feature of Fig. 4 is the anticorrelation of α_1 and α_2 for $p < p_c$; whereas α_2 rises or recovers, α_1 falls away from its doping trajectory inferred from $p > p_c$ (thin dashed line in Fig. 4A). Recent experiments on less-disordered cuprates with $p \leq 0.1$ have shown that this trend continues well into the underdoped regime with $\rho_{ab}(T)$, becoming purely T^2 at low to intermediate temperatures (*30*). The distinct momentum dependences of the two scattering processes, suggested by the ADMR experiments on Tl2201 (*18*), provides a means to interpret qualitatively the anticorrelation of α_1 and α_2 within a scenario of progressive destruction of the underlying Fermi surface (*31*). As *p* falls below p_c, sections of Fermi surface near the Brillouin zone boundary begin to disappear. Although this may not necessarily have a detrimental effect on the isotropic term α_2, for α_1 the reduction in the total number of coherent states is more than offset by the removal of the strong scattering sinks near (π, 0), leading to an overall decrease, or at best saturation, in α_1 with further reduction in carrier density.

Recent angle-resolved photoemission spectroscopy (ARPES) studies of underdoped cuprates have hinted at a gradual erosion of the Fermi arc length with decreasing temperatures below T^* (*32*). For $0.15 \leq p \leq 0.19$, T^* is about equal to $E^*/2$ and is of the order of 100 K (*33*). Because our fitting range extends up to 200 K, the changes observed in $\alpha_1(p)$ and $\alpha_2(p)$ for $p < p_c$ imply that actually quasi-particle states are lost at temperatures $T > T^*$. Moreover, given that our fitting parameters are relatively insensitive to the temperature range, we conclude that the antinodal states must play little or no role in the conductivity at any given temperature. One hint toward understanding this lies in the absolute value of α_1 (=1 μohm•cm K^{-1}) at $p = p_c$. For a 2D metal, $d\rho_{ab}/dT = (2\pi\hbar d/e^2 v_F k_F)d(1/\tau)/dT$, where *d* is the interlayer spacing (=0.64 nm in LSCO), v_F is the Fermi velocity, and k_F the Fermi wave vector. Taking typical (*p*-independent) values of v_F ($=8.0 \times 10^4$ ms^{-1}) and k_F ($=7$ nm^{-1}) for the antinodal states in overdoped LSCO (*34*), we find that at $p = p_c$ the momentum-averaged scattering rate $\hbar/\tau \sim \pi k_B T$. Given that the anisotropic scattering rate varies as $\cos^2 2\varphi$ within the plane (φ being the angle between the *k* vector and the Cu-O-Cu bond direction) (*18*), we find that $\hbar/\tau \sim 2\pi k_B T$ for states near (π, 0). This level of scattering intensity is consistent with the so-called Planckian dissipation limit (*35*) beyond which Bloch-wave propagation becomes inhibited, that is, the quasiparticle states themselves become incoherent.

The picture that emerges then is the following. As the carrier number falls and the system approaches the Mott insulating state at $p = 0$, the effective interaction responsible for the (anisotropic) *T*-linear scattering term becomes progressively stronger. Because the strength of this term is closely correlated with T_c, the same interaction also drives up both the superconducting transition temperature and the condensation energy, the latter reaching a maximum at $p = p_c$ (*33*, *36*). At this point, however, scattering intensity is so strong that those states near (π, 0) begin to de-cohere, and, given the proportionality with temperature, incoherence occurs at all finite temperatures, not just below T^*. With further reduction in doping, the strength of the interaction continues to rise, causing yet more states to lose coherence and leading to an overall reduction or saturation in $\alpha_1(p)$ and a corresponding suppression in the condensation energy (*36*) and the superfluid density (*37*).

One is then left with the intriguing possibility that the pseudogap itself forms in response to this intense scattering, the electronic ground state lowering its energy through gapping out the incoherent, highly energetic antinodal states and, in so doing, preventing scattering of the remnant quasi-particle states into those same regions. The temperature-invariant violation of the Planckian dissipation limit then provides a possible explanation as to why the pseudogap itself is a non–states-conserving gap (*3*, *33*): In contrast to a conventional (Bardeen-Cooper-Schrieffer) superconducting gap where the full density of states is recovered above the (two-particle) gap energy, pseudogapped states near (π, 0) are effectively removed at all energies up to the bandwidth (*38*) or the Coulomb energy (*39*). Along the nodal directions, where this anisotropic scattering term is absent, however (*18*), the nodal quasi-particle states remain intact (protected) at all *T* and *p* across the entire superconducting dome.

Concluding remarks. It has been a long-standing mystery why the pseudogap in cuprates vanishes at a singular doping level, p_c, beyond optimal doping. By tracking the evolution of the in-plane resistivity from the overdoped side, we have uncovered a change in the electrical transport at $p = p_c$ that appears to coincide with the onset of incoherence (at the antinodal regions), thus offering a possible explanation for its precise location in the cuprate phase diagram and the subsequent reduction in the strength of the superconductivity with lowering carrier concentration. And although Planckian dissipation is quantum criticality of sorts, it does not appear to be consistent with any simple one-parameter scaling hypothesis (*40*) in just the same way as the doping evolution of the transport coefficients is difficult to reconcile with conventional quantum criticality.

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Revealing the Maximum Strength in Nanotwinned Copper

L. Lu,[1]* X. Chen,[1] X. Huang,[2] K. Lu[1]

The strength of polycrystalline materials increases with decreasing grain size. Below a critical size, smaller grains might lead to softening, as suggested by atomistic simulations. The strongest size should arise at a transition in deformation mechanism from lattice dislocation activities to grain boundary–related processes. We investigated the maximum strength of nanotwinned copper samples with different twin thicknesses. We found that the strength increases with decreasing twin thickness, reaching a maximum at 15 nanometers, followed by a softening at smaller values that is accompanied by enhanced strain hardening and tensile ductility. The strongest twin thickness originates from a transition in the yielding mechanism from the slip transfer across twin boundaries to the activity of preexisting easy dislocation sources.

The strength of polycrystalline materials increases with decreasing grain size, as described by the well-known Hall-Petch relation (*1*, *2*). The strengthening originates from the fact that grain boundaries block the lattice dislocation motion, thereby making plastic deformation more difficult at smaller grain sizes. However, below a certain critical size, the dominating deformation mechanism may change from lattice dislocation activities to other mechanisms such as grain boundary–related processes, and softening behavior (rather than strengthening) is expected (*3*, *4*). Such a softening phenomenon has been demonstrated by atomistic simulations, and a critical grain size of maximum strength has been predicted (*5*–*7*). In pure metals, an impediment to determining the grain size that yields the highest strength is the practical difficulty of obtaining stable nanostructures with extremely small structural domains (on the order of several nanometers). The driving force for growth of nanosized grains in pure metals, originating from the high excess energy of numerous grain boundaries, becomes so large that grain growth may take place easily even at ambient temperature or below.

Coherent twin boundaries (TBs), which are defined in a face-centered cubic structure as the (111) mirror planes at which the normal stacking sequence of (111) planes is reversed, are known to be as effective as conventional grain boundaries in strengthening materials. Strengthening has been obtained in Cu when high densities of nanometer-thick twins are introduced into submicrometer-sized grains (*8*–*10*). In addition, coherent TBs are much more stable against migration (a fundamental process of coarsening) than conventional grain boundaries, as the excess energy of coherent TBs is one order of magnitude lower than that of grain boundaries. Hence, nanotwinned structures are energetically more stable than nanograined counterparts with the same chemical composition. The stable nanotwinned structure may provide samples for exploring the softening behavior with very small domain sizes. Here, we prepared nanotwinned pure Cu (nt-Cu) samples with average twin thickness ranging from a few nanometers to about 100 nm.

High-purity (99.995%) Cu foil samples composed of nanoscale twin lamellae embedded in submicrometer-sized grains were synthesized by means of pulsed electrodeposition. By increasing the deposition rate to 10 nm/s, we succeeded in refining the mean twin thickness (i.e., the mean spacing between adjacent TBs, hereafter referred to as λ) from a range of 15 to 100 nm down to a range of 4 to 10 nm (see supporting online material). The as-deposited Cu foils have an in-plane dimension of 20 mm by 10 mm and a thickness of 30 μm with a uniform microstructure. Shown in Fig. 1, A to C, are transmission electron microscopy (TEM) plane-view images of three as-deposited samples with λ values of 96 nm, 15 nm, and 4 nm, respectively. The TEM images indicate that some grains are irregular in shape, but low-magnification scanning electronic microscopy images, both cross section and plane view, show that the grains are roughly equiaxed in three dimensions. Grain size measurements showed a similar distribution and a similar average diameter of about 400 to 600 nm for all nt-Cu samples. Twins were formed in all grains (see the electron diffraction pattern in Fig. 1D), and observations of twins in a large number of individual grains revealed no obvious change in the twin density from grain to grain. Note that in all samples, the edge-on twins that formed in different grains are aligned randomly around the foil normal (growth) direction (*8*, *11*), in agreement with a strong [110] texture determined by x-ray diffraction (XRD). For each sample, twin thicknesses were measured from a large number of grains, which were detected from numerous TEM and high-resolution TEM (HRTEM) images, to generate a distribution. Figure 1E illustrates the 492 measurements for the sample with the finest twins; the majority yielded spacings between twins smaller than 10 nm, with a mean of 4 nm. For simplicity, each nt-Cu sample is identified by its mean twin thickness; for example, the sample with λ = 4 nm is referred to as nt-4.

Figure 2 shows the uniaxial tensile true stress–true strain curves for nt-Cu samples of various λ values. Also included are two stress-strain curves obtained from a coarse-grained Cu (cg-Cu) and an ultrafine-grained Cu (ufg-Cu) that has a similar grain size to that of nt-Cu samples but is free of twins within grains. Two distinct features are observed with respect to the λ dependence of the mechanical behavior of nt-Cu. The first is the occurrence of the λ giving the highest strength. All stress-strain curves of nt-Cu samples in Fig. 2, A and B, are above that of the ufg-Cu, indicating a strengthening by introducing twins into the submicrometer grains. However, such a strengthening does not show a linear relationship with λ. For λ > 15 nm (Fig. 2A), the stress-strain curves shift upward with decreasing λ, similar to the strengthening behavior reported previously in the nt-Cu (*9*, *11*) and nanocrystalline Cu (nc-Cu) (*12*–*15*) samples (Fig. 3A). However, with further de-

[1]Shenyang National Laboratory for Materials Science, Institute of Metal Research, Chinese Academy of Sciences, Shenyang 110016, P.R. China. [2]Center for Fundamental Research: Metal Structures in Four Dimensions, Materials Research Department, Risø National Laboratory for Sustainable Energy, Technical University of Denmark, DK-4000 Roskilde, Denmark.

*To whom correspondence should be addressed. E-mail: llu@imr.ac.cn

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